

University of Hertfordshire (Master in Data Science)

■ REPORT T

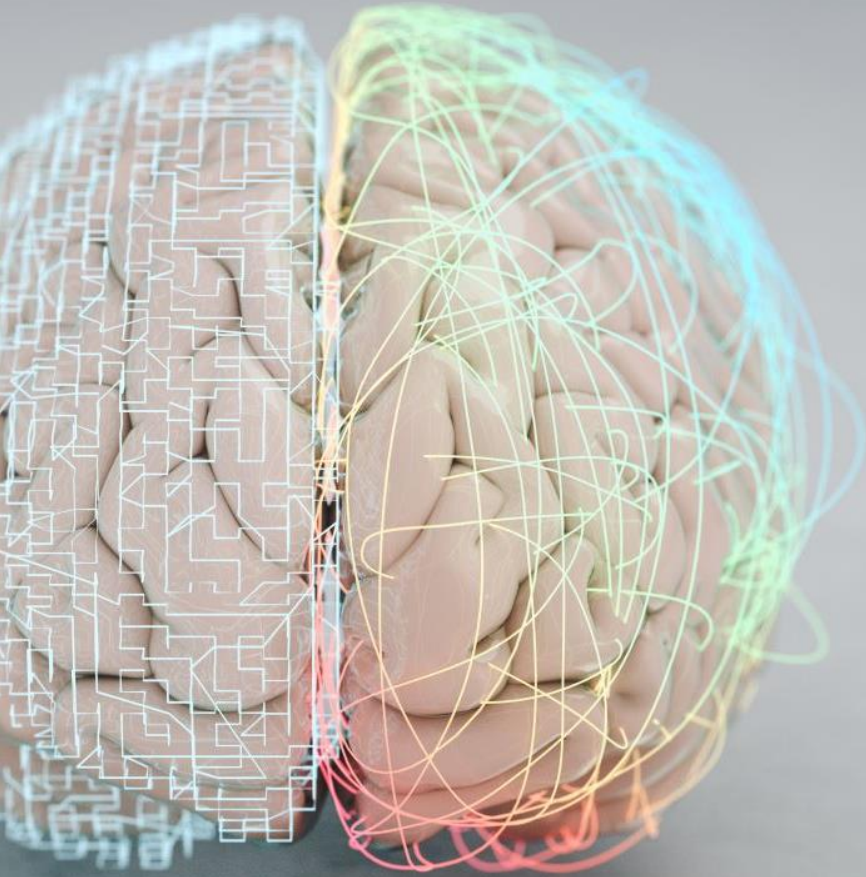
■ First Presentation

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- GitHub Link
<https://github.com/hafiz2343243/Thesis-Master-in-Data-Science-.git>

A close-up photograph of a blue and silver ballpoint pen lying diagonally across a document. The document features a bar chart with several blue bars of varying heights. The background is a light cream color.

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Aim of the study

- To develop an advanced deep-learning technique to classify brain tumors effectively.
- Implementation of transfer learning models along with the concatenation of various transfer learning models to verify the results and efficiency by considering the performance evaluation metrics.
- Introducing a novel proposed deep convolutional neural network to get higher accuracy with fewer parameters.
- Improving the overall accuracy of the used approaches like preprocessing and augmentation in detecting and classifying brain tumors.
- Comparing the performance of the proposed methodology with various transfer learning models through performance evaluation metrics, training and testing curves and confusion matrices.

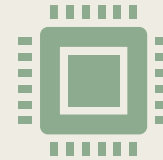
Contributions



The deployed transfer learning algorithms, such as EfficientNetB6, VGG16, XceptionNet, ResNet101, and ResNet50 whereas the InceptionNet+VGG19 and EfficientNetV2+Densenet121 will be tested for Brain tumor classification effectiveness.



In order to compete with the TL models, a novel DCNN model will be constructed from scratch with fewer parameters and improved efficiency and accuracy.



The lesser the number of parameters, lower the computing cost.



The performance of all deployed TL models and proposed DCNN model tested not only with accuracy but also with additional performance assessment metrics such as recall, precision, f1 score, and validation loss.



Research Gap

1. Research Gaps in AI for Brain Cancer Detection:

- Significant knowledge gaps must be addressed to improve AI's ability to identify brain cancer.
- Research should focus on understanding AI's specific capabilities and developing strategies for advancement.

3. Challenges with Medical Image Datasets:

- Medical image datasets, particularly in the MRI domain, are challenging to handle due to their complexity.
- Conventional machine learning classifiers struggle with these datasets, while deep learning models, such as convolutional neural networks (CNNs), perform better.

4. Need for Advanced Models:

- A model capable of significant progress in MRI dataset analysis is required.

5. Importance of Accuracy, Sensitivity, and Specificity:

- Addressing research gaps necessitates attention to model accuracy, along with sensitivity and specificity, to ensure reliable performance.

6. Model Competence and Credibility:

- The model should exhibit high credibility and competence across a broad range of tasks to effectively address the challenges.

Introduction

Brain cancer is an infrequent and particularly deadly malignancy of very meager incidence, as new studies have revealed [1].

Because of the detailed anatomy and complicated brain physiology, diagnosing and treating this condition is very difficult. Surgically and chemotherapeutically, addressing this issue is not without its risks of brain damage [2].

The latest available data on the Global Cancer Statistics also confirm that brain and central nervous system cancers are common. The most frequent neoplasms are located in the brain and spinal cord, among which Astrocytoma, Oligodendrogliomas represent the majority [3].

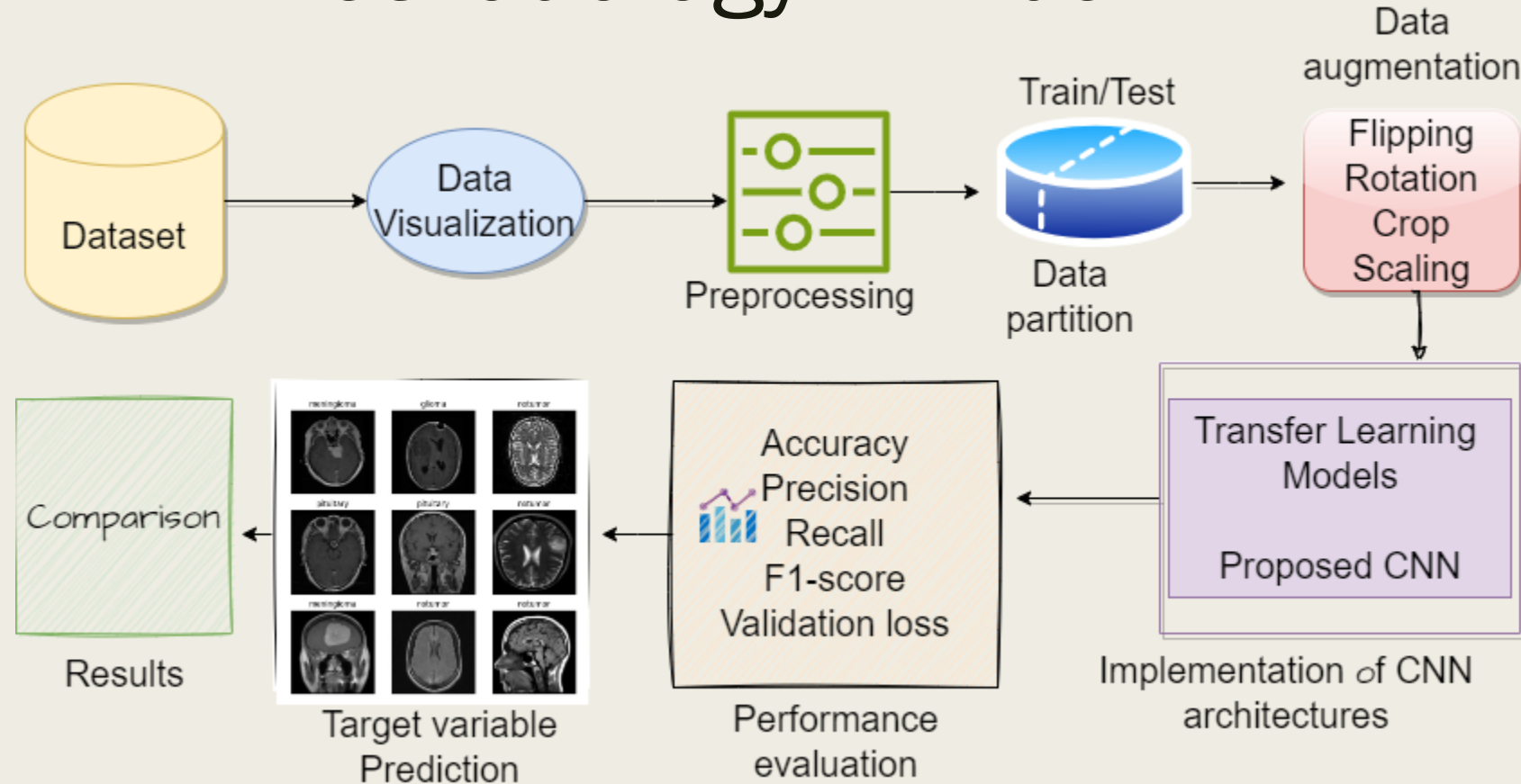
The record shows there were 251,329 cancer fatalities and 308,102 fresh encephalon/CNS cases in 2020. The leading cause of mortality in the world since time immemorial continues to be the brain and central nervous system [4].

In 2019, there were 347,992 new brain cancers around the world. A total of 138,605 were men and 107,648 women [3], [5].

Literature Review

Year [Ref]	Dataset	Methodology	Results	Limitation
			Accuracy	
2024[6]	BraTs2020	SHAP + proposed ensemble	99.15%	Misclassification + high inference time
2022 [7]	28413, BraTs2018	ResNet101+Feature selection + PCA	94.4%	High computational time
2022 [8]	Figshare & 28413, BraTs2018	MANet	96.51% & 94.91%	Lack of multitasking classification
2022 [9]	3064, FigShare 2017	Bayesian Optimized CNN	98.70%	Data Imbalance and less hyperparameter tuning
2023 [10]	253, files Kaggle	HHOCNN	98%	Lack of optimized feature selection
2023 [11]	3064, FigShare 2017	BTC FCNN	98.86%	Less number of training iteration
2023[12]	8487, MRI	AlexNet + Data Augmentation	98.24%	Unstable Characteristics

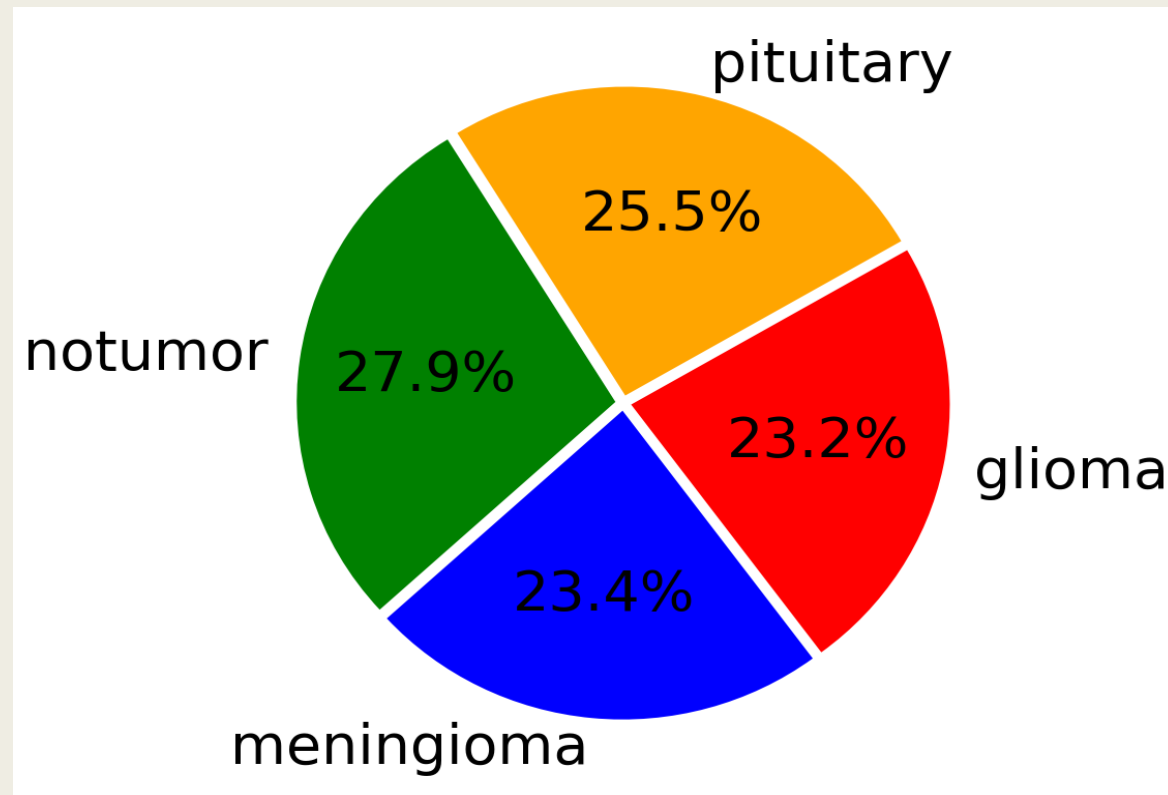
General Block Diagram of the Proposed Methodology will be...



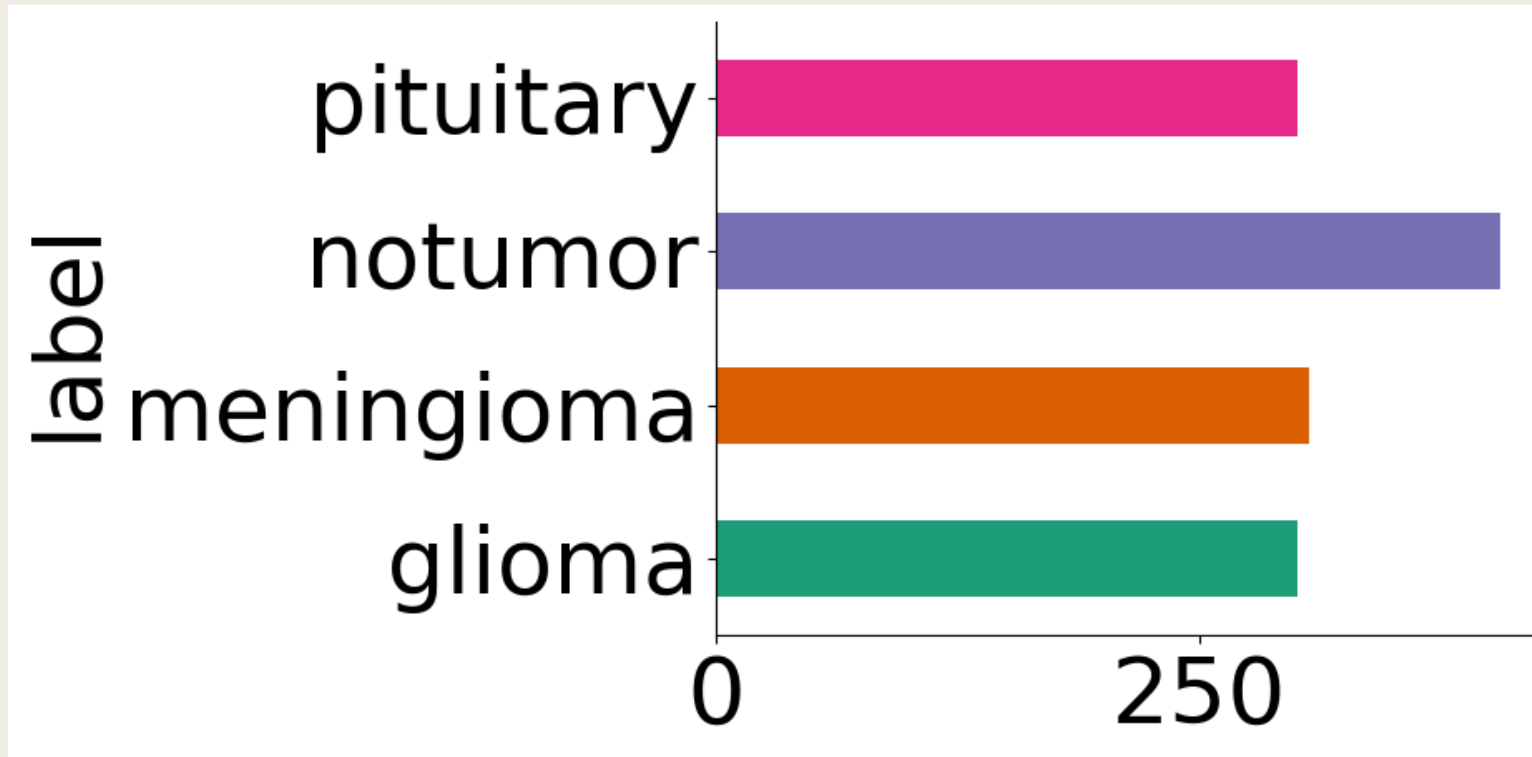
Used Data Set

- An open-source dataset used
- This dataset is a combination of three open source datasets Figshare+SARTAJ + Br35H
- The link of the dataset is here : <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>
- Total Four classes pituitary, Glioma, Meningioma, Notumor
- Total 7023 sample, training samples consisted of a total of 5712 MRI pictures.

Target Class distribution of the Brain Tumor Dataset

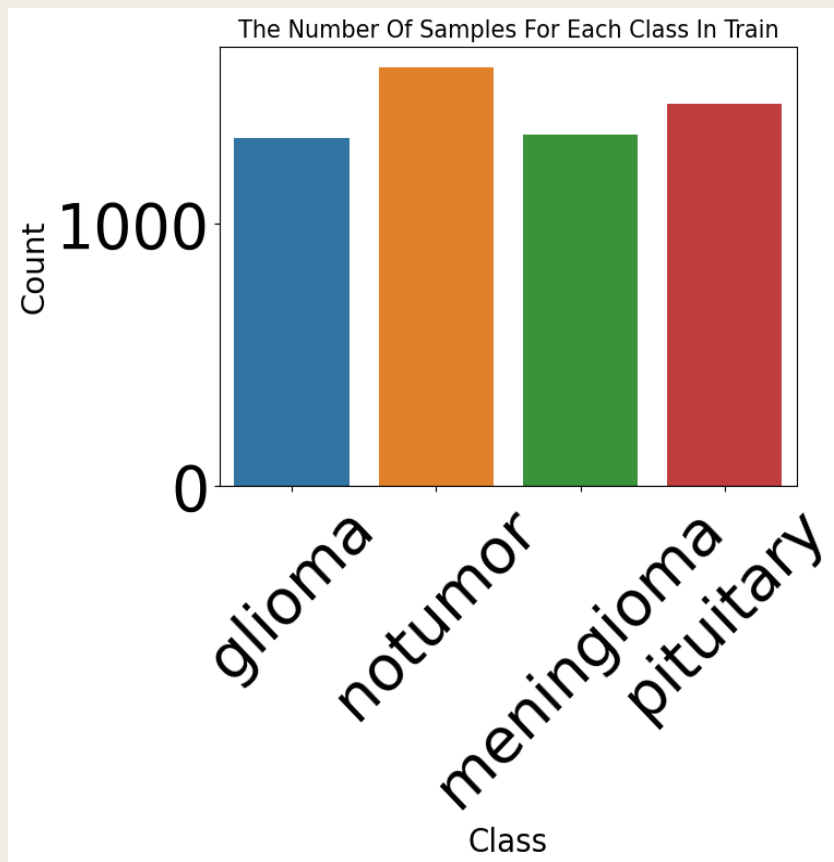


Brain Tumor Dataset Test samples



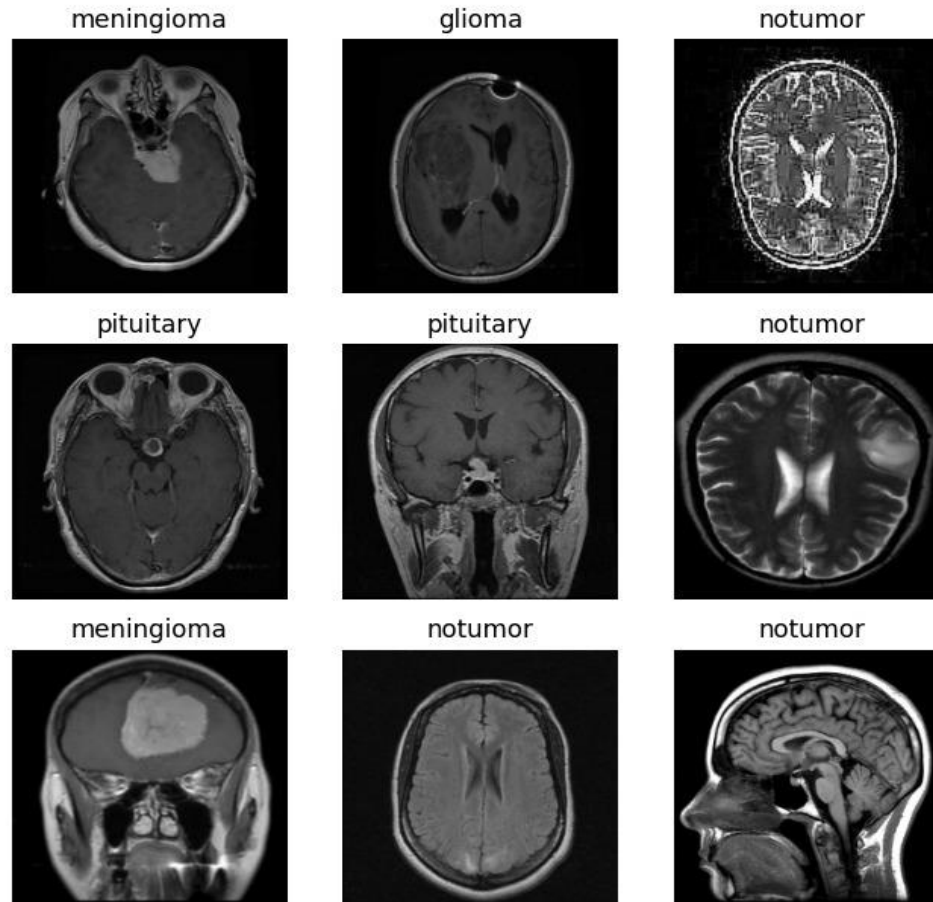
Classes	Total testing samples
Pituitary	300
No tumor	405
Meningioma	306
Glioma	300

Brain Tumor Dataset Training samples



Classes	Total training samples
Pituitary	1457
No tumor	1595
Meningioma	1339
Glioma	1321

Samples from the Dataset



Data Ethics

- Data is open source
- No ethical concern is raised to the personal data usage and privacy
- Publicly available for the research.
- Project focus on the data collection, preprocessing techniques, data visualization, data augmentation approaches, implementation of the transfer learning and the proposed DCNN models, implementation of the performance evaluation metrics for the comparison with existing studies.
- Testing methods focused on the standardized procedures, like confusion matrix, accuracy, precision, recall , f1score etc.
- Ethical Approval is not required as the dataset is publicly available.
- The purpose of this study is for the academic research only, ensuring minimal ethical concerns.

Data Management Plan

- Data Collection will be done by downloading the data in the form of images with four folders for each class including testing and training samples.
- Backup of the data will be placed with weekly local backups and cloud backups bi-weekly.
- Version control will be managed via github with weekly commits.
- Repository organization will be done with structured folders with code, documentation and results.
- Goals will be to ensure the safe storage, proper Git version control, smooth and organized work flow through out the project.

Time	WEEK1	WEEK2	WEEK3	WEEK4	WEEK5	WEEK6	WEEK7	WEEK8	WEEK9	WEEK10	WEEK11	WEEK12	WEEK13	WEEK14
Project selection														
Choosing project	Done													
Supervisor allocation	Done													
Related work & Preprocessing														
Literature review														
Dataset download & Preprocessing														
		Ongoing.....												
		Work in progress												
PDM plan														
Ppt & Report			In progress.....											
Methodology implement														
Coding + model s evaluation			Upcoming in future.....											
Test & Evaluation +Writing														
Results										Upcoming in future...				
Writing Draft												Upcoming.....		
Final Review & Submission												Upcoming....		

Data Augmentation strategies

Data augmentation strategy	Value of the parameter
Rotation	30
Image Width Shift	0.20
Image Height Shift	0.25
Shear Range	0.20
Shear range	0.22
Horizontal flip	True
Fill mode	Nearest
Cropping	0.1

Performance Evaluation Metrics

Precision: The above statement clarifies the correct identification of predicted cases, particularly those categorized as True Positives, within the whole set of Positive predictions, which includes both True and False Positives. The following is a mathematical representation of the idea under consideration.

$$Precision_{(pr)} = \frac{T_P}{T_P + F_P} \times 100\% \quad (1)$$

Accuracy: The measure accuracy refers to the proportion of correctly categorized cases in comparison to the total number of accurate and incorrect examples. The following is the mathematical expression:

$$Accuracy_{(AC)} = \frac{T_P + T_N}{Total\ number\ of\ samples} \times 100\% \quad (2)$$

F1-Score : The metric under consideration quantifies the balance between precision and recall. The harmonic mean of accuracy and recall is being referred to. When confronted with the challenge of class-imbalanced data, it is worth noting that the F1-Score metric is quite valuable. The calculation is presented in the following manner.

$$F1 - Score_{(F1S)} = 2 \times \frac{PRE \times REC}{PRE + REC} \times 100\% \quad (3)$$

Recall : Upon evaluating the model's efficacy in mitigating the occurrence of False Negatives, it is imperative to delve into a comprehensive analysis of its capabilities. The metric denoted as Recall quantifies the ratio between True Positives and the sum of False Positives and True Positives. Mathematically, recall can be defined as the ratio of true positive instances to the sum of true positive and false negative instances:

$$Recall_{(RE)} = \frac{T_P}{T_P + F_N} \times 100\% \quad (4)$$

Validation loss: The validation set is a subset of the dataset put aside to test the model's performance. The validation loss, like the training loss, is determined by adding the errors for each sample in the validation set.

$$Loss\ function = \ln \sum N \text{if}(\hat{y}_i, y_i)$$

References

- [1] Brain Tumor - Statistics. (2023, May 31). Cancer.Net. <https://www.cancer.net/cancer-types/brain-tumor/statistics>
- [2] Z. Akkus, A. Galimzianova, A. Hoogi, D. L. Rubin, and B. J. Erickson, “Deep Learning for Brain MRI Segmentation: State of the Art and Future Directions,” *Journal of Digital Imaging*, vol. 30, no. 4, pp. 449–459, Jun. 2017, doi: 10.1007/s10278-017-9983-4.
- [3] Kamnitsas, Konstantinos, Christian Ledig, Virginia FJ Newcombe, Joanna P. Simpson, Andrew D. Kane, David K. Menon, Daniel Rueckert, and Ben Glocker. 2017. “Efficient Multi-Scale 3D CNN with Fully Connected CRF for Accurate Brain Lesion Segmentation.” *Medical Image Analysis* 36:61–78.
- [4] O. Özkaraca *et al.*, “Multiple Brain Tumor Classification with Dense CNN Architecture Using Brain MRI Images,” *Life*, vol. 13, no. 2, 2023, doi: 10.3390/life13020349.
- [5] N. Ullah, M. S. Khan, J. A. Khan, A. Choi, and M. S. Anwar, “A Robust End-to-End Deep Learning-Based Approach for Effective and Reliable BTD Using MR Images,” *Sensors*, vol. 22, no. 19, 2022, doi: 10.3390/s22197575.
- [6] V. S. Yilmaz, M. Akdag, Y. Dalveren, R. O. Doruk, A. Kara, and A. Soyly, “Investigating the Impact of Two Major Programming Environments on the Accuracy of Deep Learning-Based Glioma Detection from MRI Images,” *Diagnostics*, vol. 13, no. 4, 2023, doi: 10.3390/diagnostics13040651.
- [7] N. A. Samee *et al.*, “Classification Framework for Medical Diagnosis of Brain Tumor with an Effective Hybrid Transfer Learning Model,” *Diagnostics*, vol. 12, no. 10, 2022, doi: 10.3390/diagnostics12102541.